Algebraic Reasoning & Properties — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Name the Property

Watch Me — Worked Example

"If $2x = 12$, then $x = 6$." Both sides were divided by 2. **Answer: Division Property of Equality.**
"If $m\angle A = 40°$, then $40° = m\angle A$." The equation is flipped. **Answer: Symmetric Property.**

1. If $x + 7 = 12$, then $x = 5$.

Subtraction Prop. of Equality

2. If $\frac{x}{3} = 4$, then $x = 12$.

Multiplication Prop. of Equality

3. $m\angle K = m\angle K$

Reflexive Property

4. $5(x - 2) = 5x - 10$

Distributive Property

5. If $AB = 8$ and $AB + BC = 15$, then $8 + BC = 15$.

Substitution Property

6. If $PQ = RS$ and $RS = TU$, then $PQ = TU$.

Transitive Property

Type 2 Solve and Give a Reason for Each Step

Watch Me — Worked Example

$4x - 3 = 2x + 9$	Given
$2x - 3 = 9$	Subtraction Property of Equality (subtract $2x$)
$2x = 12$	Addition Property of Equality (add 3)
$x = 6$	Division Property of Equality (divide by 2)

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7. Solve $3x + 5 = 20$. Give a reason for each step.

$3x + 5 = 20$	Given
$3x =$ 15	**Subtraction Property of Equality**
$x =$ 5	**Division Property of Equality**

8. Solve $2(x - 4) = 3x - 1$. Give a reason for each step.

$2(x - 4) = 3x - 1$	Given
$2x - 8 = 3x - 1$	**Distributive Property**
$-8 = x - 1$	**Subtraction Property of Equality**
$-7 = x$	**Addition Property of Equality**
$x = -7$	**Symmetric Property**

Type 3 Properties With Segments and Angles

Watch Me — Worked Example

Given $m\angle 1 + m\angle 2 = 90°$ and $m\angle 1 = 35°$.
$35° + m\angle 2 = 90°$ (Substitution), so $m\angle 2 = 55°$ (Subtraction Property of Equality).

9. $AB + BC = AC$, $AB = 2x$, $BC = 3x + 1$, $AC = 21$. Find x; name the property used to replace AB and BC.

$x = 4$; **Substitution**

10. $m\angle P + m\angle Q = 180°$, $m\angle P = 3x$, $m\angle Q = 2x + 30$. Find $m\angle P$.

$x = 30$, $m\angle P = 90°$

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Type 4 Which Property? Reflexive, Symmetric or Transitive

Watch Me — Worked Example

Reflexive uses **one** thing: $\overline{AB} \cong \overline{AB}$. **Symmetric** flips **two**: $\angle 1 \cong \angle 2 \Rightarrow \angle 2 \cong \angle 1$. **Transitive** chains **three**.

11. $\overline{XY} \cong \overline{XY}$

Reflexive

12. If $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.

Transitive

13. If $\overline{JK} \cong \overline{LM}$, then $\overline{LM} \cong \overline{JK}$.

Symmetric

14. HARD Is "is taller than" transitive? Is it symmetric?

transitive yes; symmetric no

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